

21 October Deep learning.

Keras

What is deep learning.

Classical ml has limitation

can't handle huge amount of data
or unstructured data

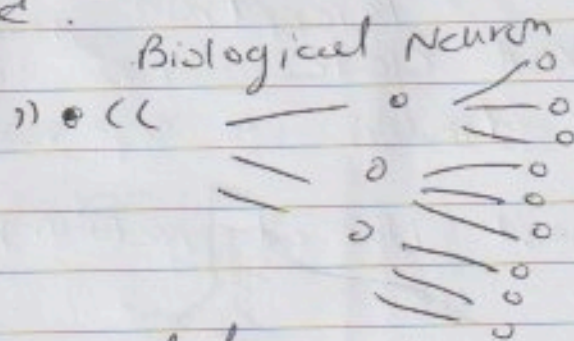
image
video
audio

problem with complex data.

What is Deep learning.

Neuron or Perceptron

Around 1991 or 1995 this concept was introduced.

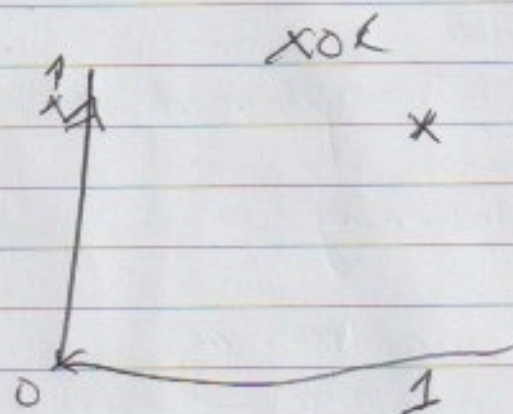
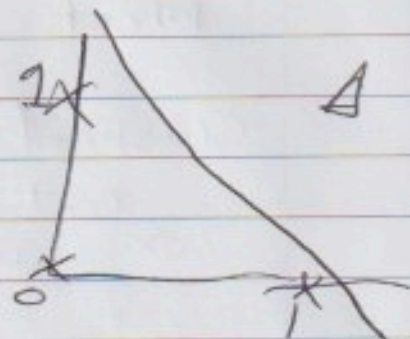
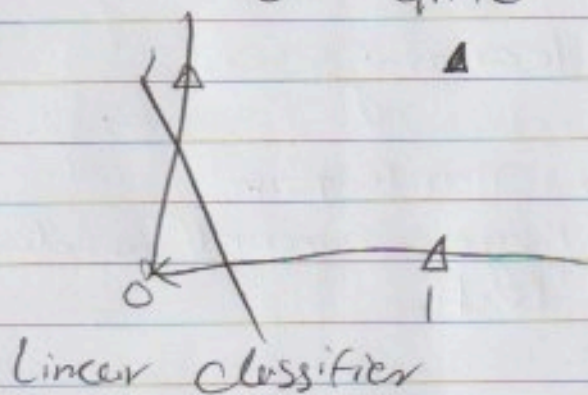


First perceptron model was intro

A	B	A OR B	A AND B	A XOR B
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Applied Perceptron on OR Gate

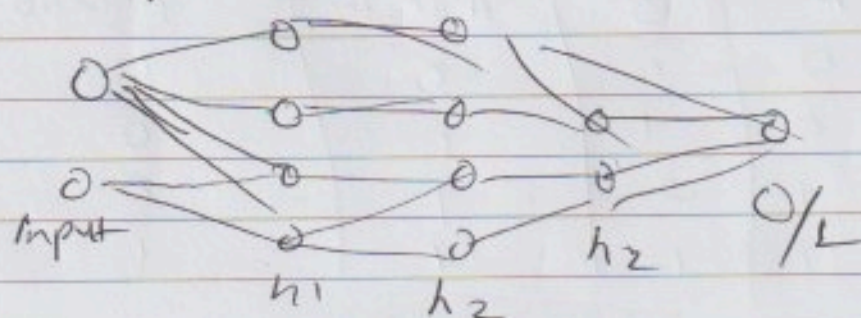
OR GATE



Perceptron model ^(?) could not solve XOR gate. as linear classifier line can be drawn to separate x from Δ

First Neural Network was Perceptron.

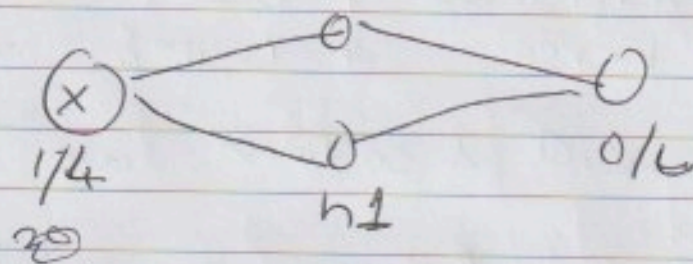
Then after 5 years of research they came up with NN.



Multiple perceptron unneuted

X	target
20	min
28	max
42	max
24	min

input features $\rightarrow$
 1 hidden layer $\rightarrow$ 2 neurons
 1 output neuron



Activation function

loss:

if it is regression mse

if output is binary
 if target is ? Binary class or not
 if it is ? Mult -

Mean Square Error
 Input: 20
 Output: 2.2

What is an Activation function

$$y = m \cdot x + c$$

$$0.1 \times 20 + 0.2 = 2.2$$

$$-0.2 \times 20 + 0.1 = -3.9$$

20 (x) $\begin{cases} 0.1 & 0.2 \\ -0.2 & -0.1 \end{cases}$ $\begin{cases} 2.2 \\ -3.9 \end{cases}$ $\begin{cases} 0.4 & 0.1 \end{cases}$ $\begin{cases} 2.2 \\ -3.9 \end{cases}$

$m(20) + c$

need to find m & c. In NN we start with random values

$$\text{Sigmoid function} = \frac{1}{1 + e^{-x}}$$

$$= \frac{1}{1 + e^{-2.2 - 3.9}}$$

if the result is not expected we go back (Forward pass & Backward pass)

Regression

Classification

target 2 classes binary
↳ or more than 2 ~~mult~~ Multic

binary $\rightarrow$ sigmoid.

multiclass $\rightarrow$ softmax.

setosa + 0.9. — has highest prob
vermical + 0.001
versa + 0.008

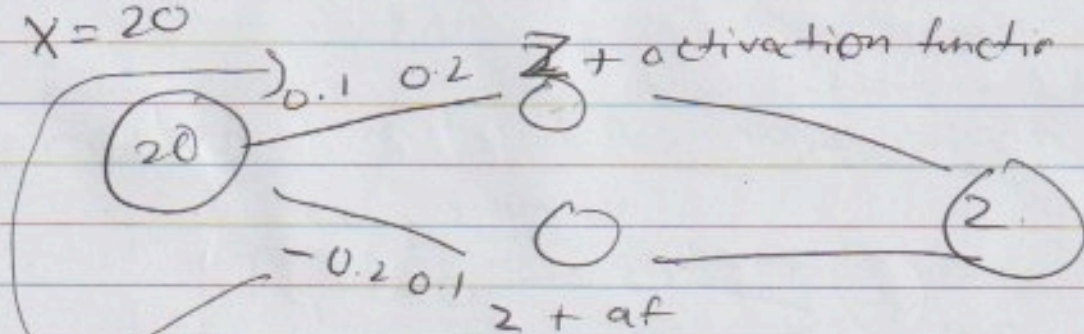
so sigmoid & softmax are activation functions.

There are different kinds of Activation function.

Sigmoid func

$$= \frac{1}{1 + e^{-x}}$$

$$x = 20$$



$$z = 20 \times 0.1 + 0.2 = 2.2$$

$$= \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-2.2}} = \dots$$

$$\rightarrow z = -0.2 \times 20 \times 0.1$$

$$= \frac{1}{1 + e^{+3.9}}$$

Get Random weights
 Do prediction
 Calculate loss
 if loss is high.
 then update weight
 and calculate.

All NN are linear lines.

Based on target we use activation func
 if the output is binary.

is multiclass

↳ categorical Cross entropy

IRIS Data.

SL	0	0	0	0
SW	0	0	0	0
PL	0	0	0	0
PW	0	0	0	0
	0	0	0	0
	0	0	0	0
	0	0	0	0

See video for explanation G. S. Ar

~~1/6~~

Keras

- ① sequential Model
- ② Functional Model
- ③ Model

1/4 - 4

$h_1 = 8$

$h_2 = 10$

$h_3 = 10$

o/l - 3

model = sequential()

```
model.add(Dense(8, input_dim=4, activation='relu'))  
model.add(Dense(10, activation='relu'))  
model.add(Dense(10, activation='relu'))  
model.add(Dense(3, activation='softmax'))
```

if the accuracy is not good then we change the number of neurons & possible hidden layers.

1

2

3

How or

Transformer.

In 2017 there is a paper published
All about attention

Keras - longer time
PyTorch.

Q1, Q2, Q3 Journals

↳ Research.

- Kegel for data sample.